

Tanu Chaudhary

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SUMMARY

Software Engineer skilled in Java, Python, Spring Boot, FastAPI, MERN stack, and Machine Learning with strong knowledge of DSA, OOPs, DBMS, and system design. Experienced in building scalable web applications, REST APIs, and AI-based solutions.

EDUCATION

Vellore Institute of Technology, Chennai

Master of Computer Applications | CGPA: 8.62

Chennai, Tamil Nadu

Expected Graduation 2027

Galgotias University, Noida, UP

Bachelor of Computer Applications (AIML) | CGPA: 8.6

2022-2025

EXPERIENCE

Google Generative AI Virtual Intern | AICTE

07/2024 – 09/2024

- Applied prompt engineering for real-time intelligent responses and automation
- Developed foundational skills in generative AI and AI automation concepts.

PROJECTS

Human Suspicious Activity Recognition System | SigLIP 2, Mamba, Deep Learning, Computer Vision, Transformers

- Developed a spatial-temporal deep learning system for suspicious activity detection and video classification.
- Achieved 97% accuracy on real-world video datasets for suspicious activity detection and classification.

Smart City Management & Complaint Resolution Platform | MERN Stack (MongoDB, Express.js, React.js, Node.js)

- Built a scalable MERN-based platform for real-time complaint tracking and civic service management.
- Developed RESTful APIs and optimized backend workflows for efficient concurrent request handling and system scalability

Crime Prediction System | XGBoost, Random Forest and Decision Tree

- Developed an ML-based crime forecasting system analyzing 50K+ crime records to predict hotspots with 89% F1-score.
- Integrated machine learning models with scalable backend services, improving prediction processing efficiency by 25%.

AI Desktop Assistant | OpenCV, Gemini API, Speech Recognition, TTS, REST APIs, and Python

- Built a system with face authentication, task automation, LLM-based intelligent responses using OpenCV and Gemini API.
- Automated 20+ tasks including app launching, web search, and system control with context-aware interactions.

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, JavaScript, C, PHP

Frontend Development: React.js, Angular, HTML, CSS, AJAX, jQuery

Backend & Systems: Node.js, Spring Boot, FastAPI, Express.js, REST APIs, Distributed Systems, Microservices, Concurrency, Multi-threading, System Design, MySQL, MongoDB

Core Computer Science: Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks

Machine Learning & AI: Machine Learning, Deep Learning, Data Preprocessing, Model Evaluation, Transformers

Tools & Platforms: Git, GitHub, Docker, AWS (EC2, S3), OpenStack, Postman, VS Code

RESEARCH & ACHIEVEMENTS

- Research: Quantum-Inspired Multi-Agent Learning for Blue Carbon Modeling (Under Review – Springer)
- Smart India Hackathon: Pre-Qualifier (2023, 2024)

CERTIFICATES

Web Development Fundamentals – IBM [Link](#)

Database Programming with SQL – Oracle [Link](#)

Microsoft AI & ML Engineering – Microsoft [Link](#)

AWS Machine Learning Foundations – AWS [Link](#)